US 20250281198A1

(19) **United States**

(12) **Patent Application Publication**

Fagan et al.

(10) **Pub. No.: US 2025/0281198 A1**

(43) **Pub. Date: Sep. 11, 2025**

(54) **ARTICULATING ULTRASONIC SURGICAL INSTRUMENTS AND SYSTEMS**

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(21) Appl. No.: **19/217,602**

(22) Filed: **May 23, 2025**

Related U.S. Application Data

(63) Continuation of application No. 17/797,175, filed on Aug. 3, 2022, filed as application No. PCT/US2021/020473 on Mar. 2, 2021, now Pat. No. 12,310,615.

(60) Provisional application No. 63/003,985, filed on Apr. 2, 2020.

Publication Classification

(51) **Int. Cl.**

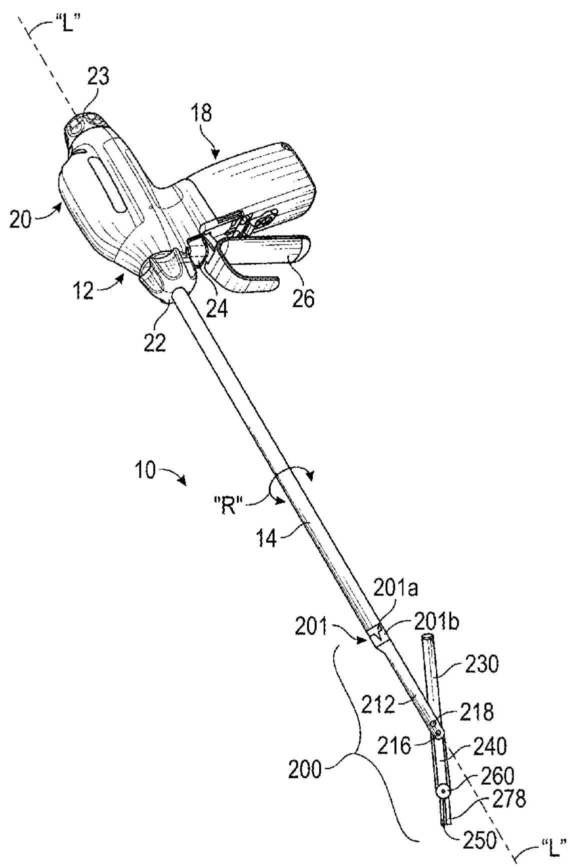
A61B 17/32	(2006.01)
A61B 17/00	(2006.01)
A61B 17/29	(2006.01)
A61B 34/37	(2016.01)

(52) **U.S. Cl.**

CPC ***A61B 17/320092*** (2013.01); ***A61B 34/37*** (2016.02); *A61B 2017/00402* (2013.01); *A61B 2017/2927* (2013.01); *A61B 2017/320071* (2017.08); *A61B 2017/320075* (2017.08); *A61B 2017/320094* (2017.08); *A61B 2017/320098* (2017.08)

(57) **ABSTRACT**

An articulating ultrasonic surgical end effector for use with a hand-held surgical instrument or a robotic surgical system includes an articulation assembly, a clevis operably coupled to the articulation assembly, and a transducer assembly pivotably coupled to the clevis. The transducer assembly includes a transducer housing and an ultrasonic transducer and a waveguide disposed within the transducer housing. The waveguide is coupled to the ultrasonic transducer and extends distally from the ultrasonic transducer. The transducer assembly also includes an ultrasonic blade disposed at the distal end of the waveguide and extending from the transducer housing and a clamp arm pivotably coupled to the transducer housing and movable relative to the ultrasonic blade between an open position and a clamping position. Ultrasonic energy produced by the ultrasonic transducer is transmitted along the waveguide to the ultrasonic blade for treating tissue.



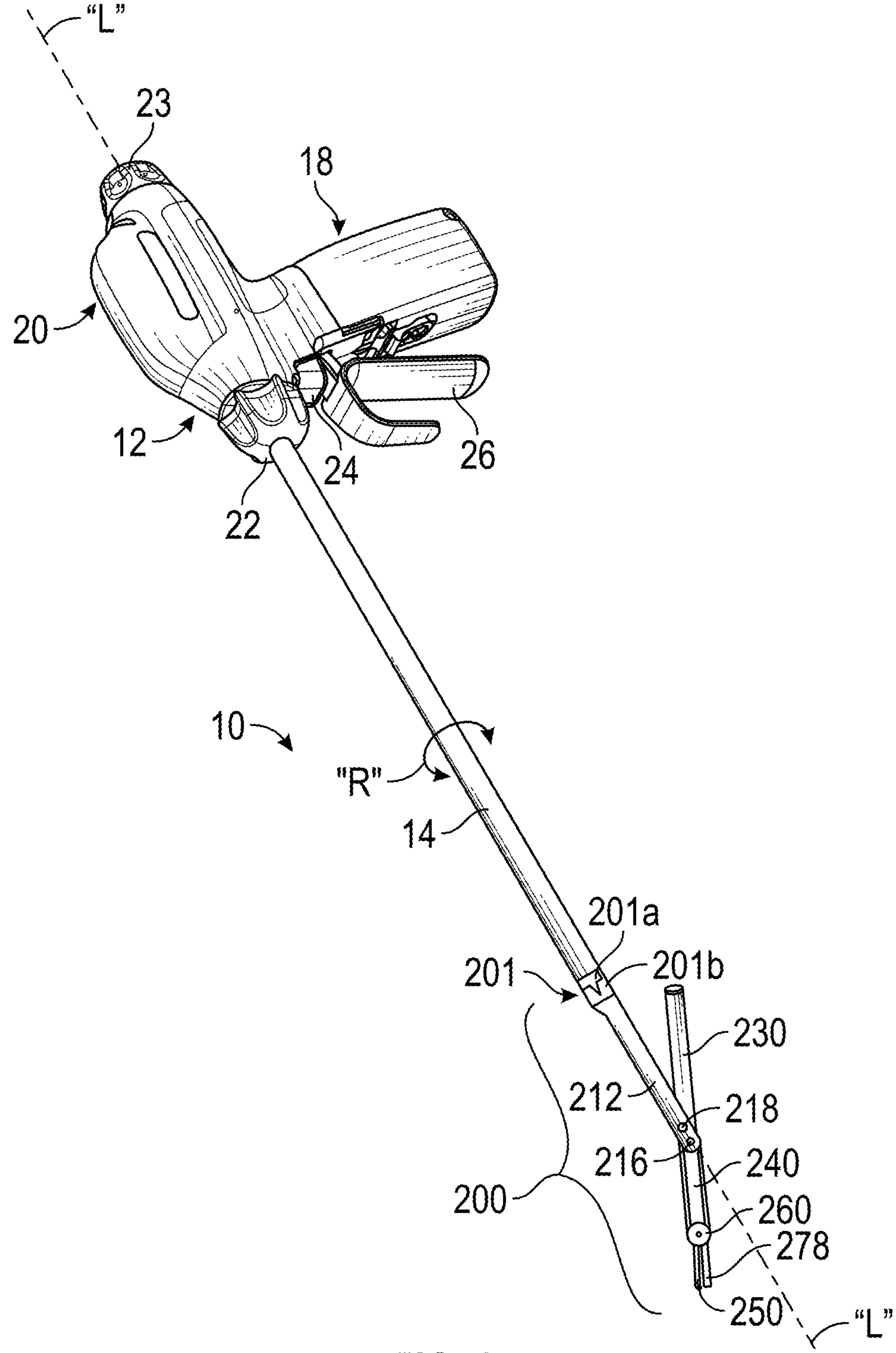

"L"
23
18
20
12
24
26
22
10
"R"
14
201a
201
201b
230
212
218
216
240
200
260
278
250
"L"

FIG. 1

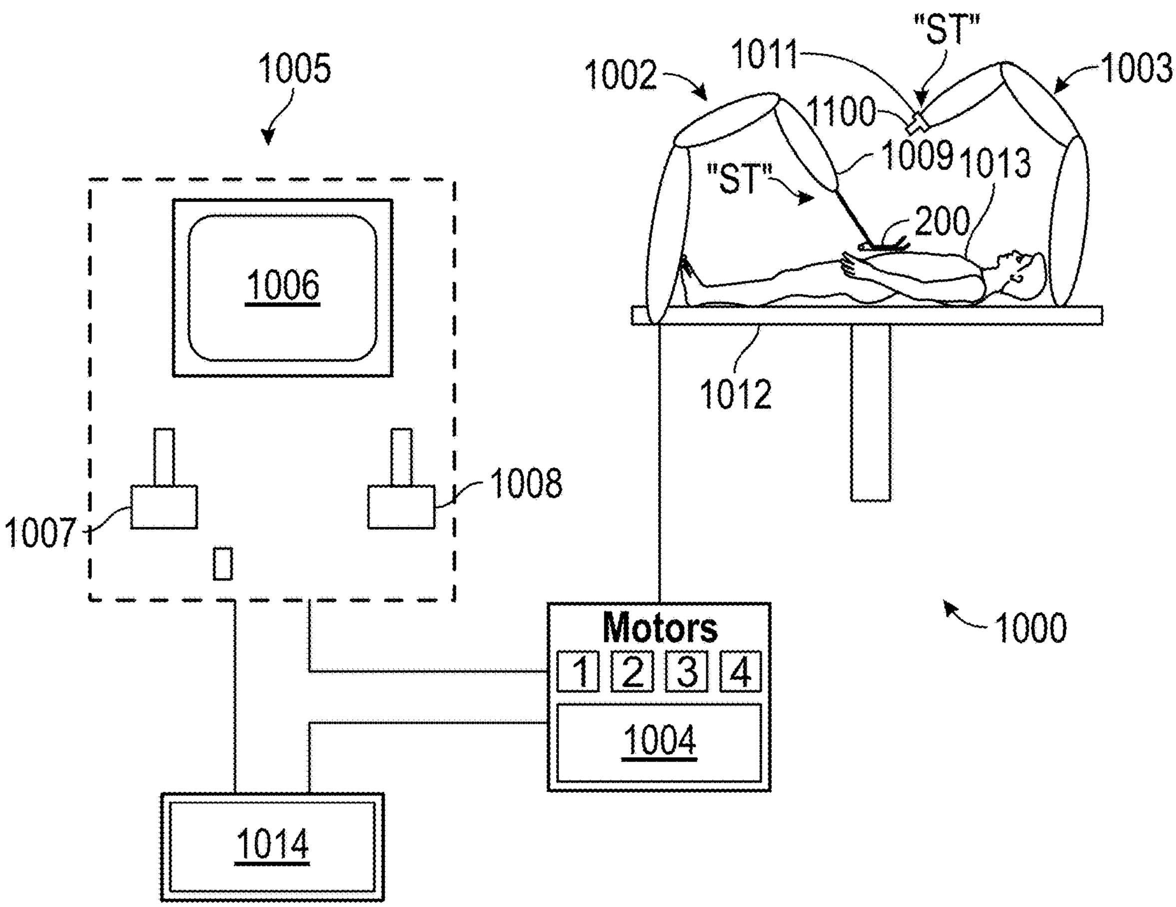

1005
1006
1007
1008
1011
"ST"
1002
1100
1003
1009
1013
"ST"
200
1012
Motors
1
2
3
4
1004
1000
1014

FIG. 2

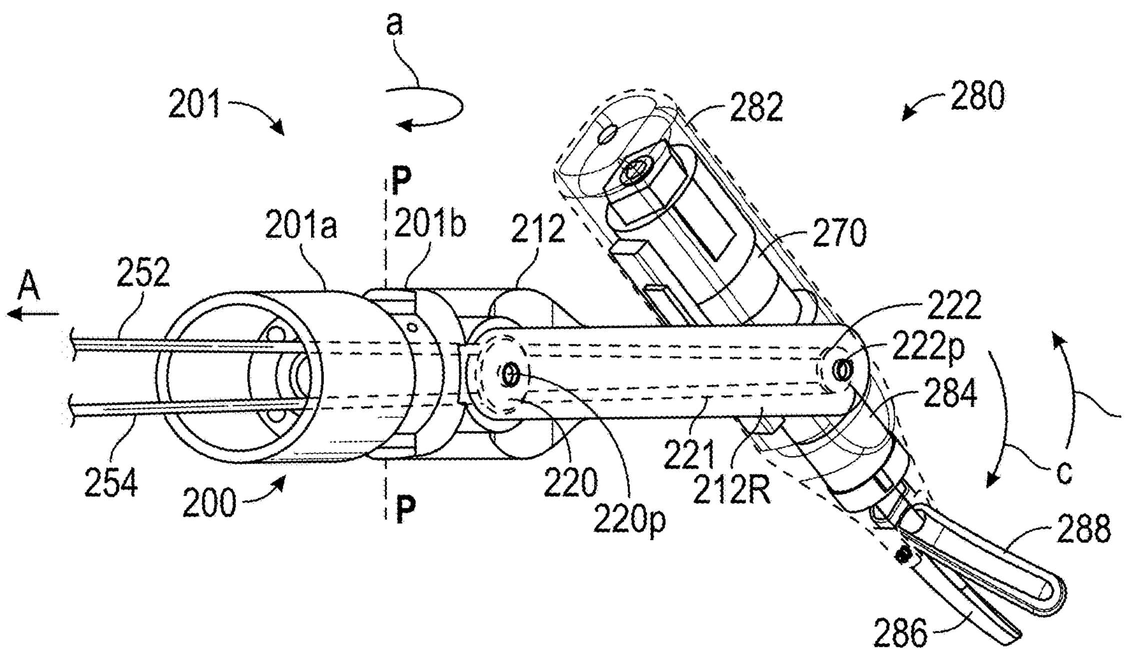

a
201
280
282
P
201a
201b
212
270
A
252
222
222p
284
b
c
254
221
200
220
212R
P
220p
288
286

FIG. 3

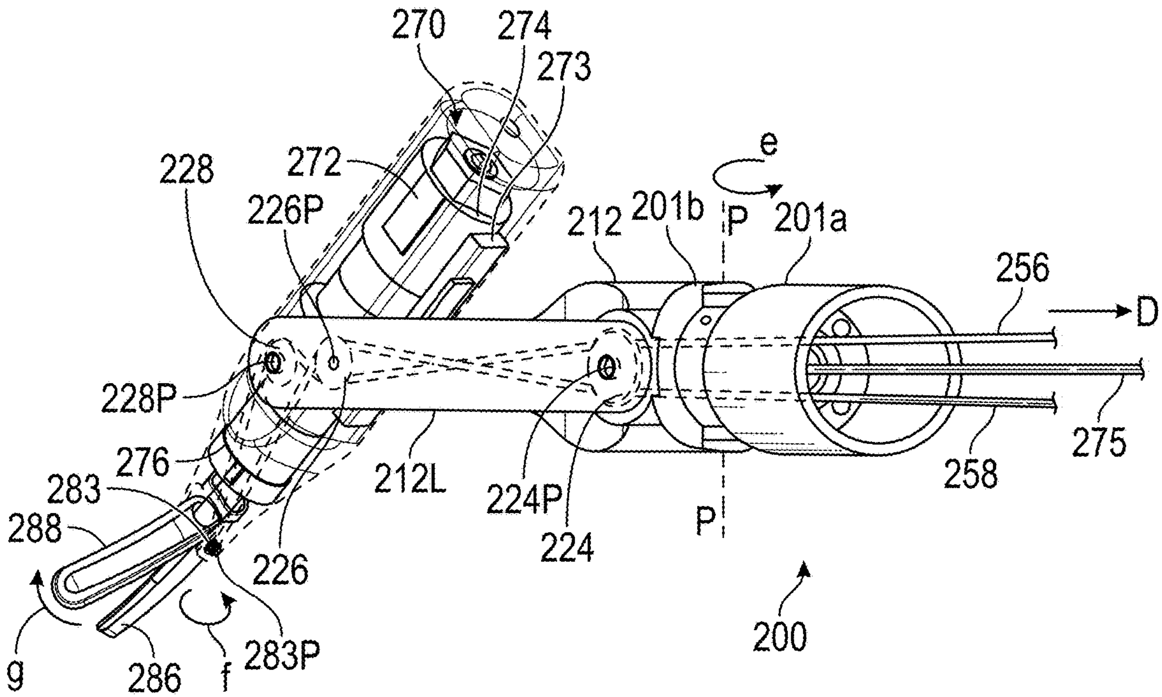
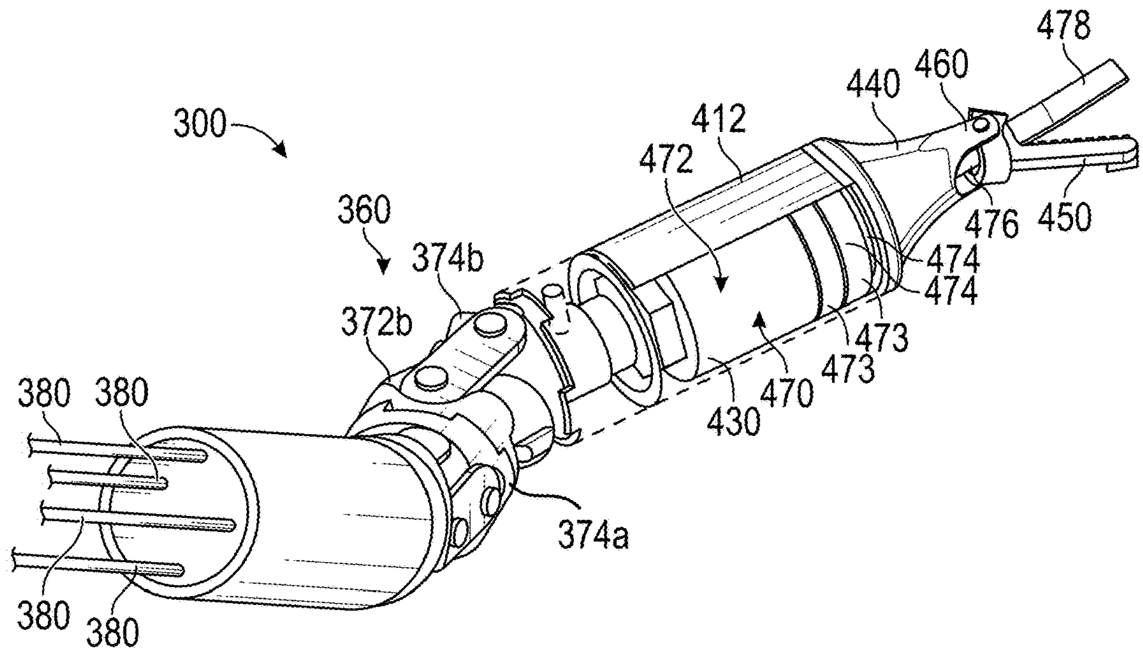
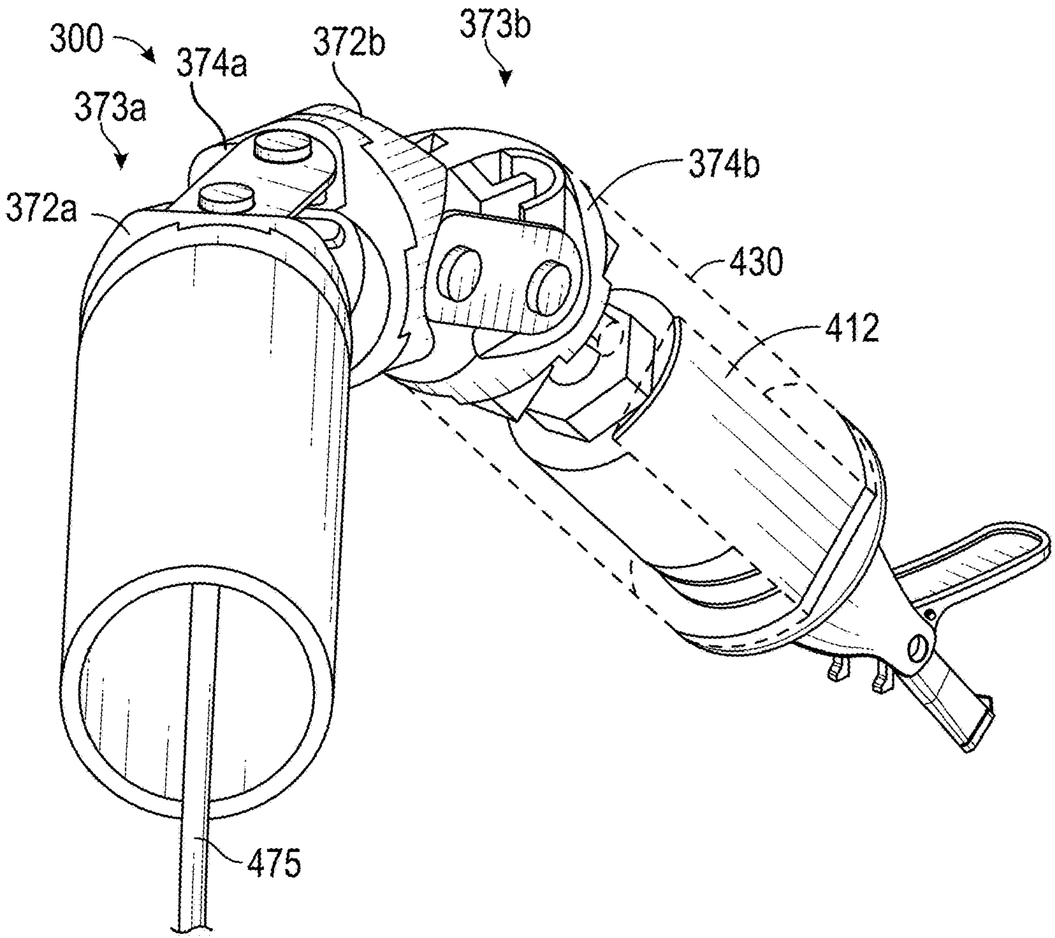

270
274
273
272
e
228
226P
212
201b
P
201a
256
D
228P
275
276
212L
258
283
224P
288
P
224
226
g
286
f
283P
200

FIG. 4



300
360
374b
372b
412
472
440
460
478
476
450
474
474
473
473
470
430
380
380
380
380
374a

FIG. 5



300
374a
372b
373b
373a
372a
374b
430
412
475

FIG. 6

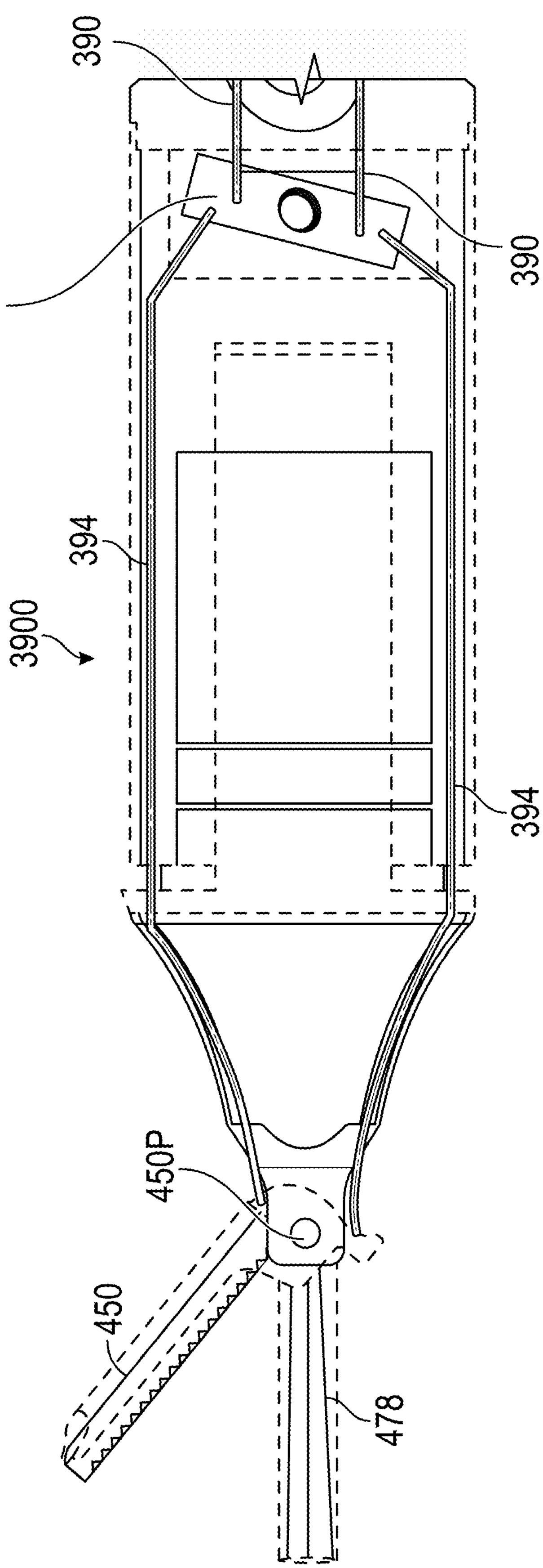


FIG. 7

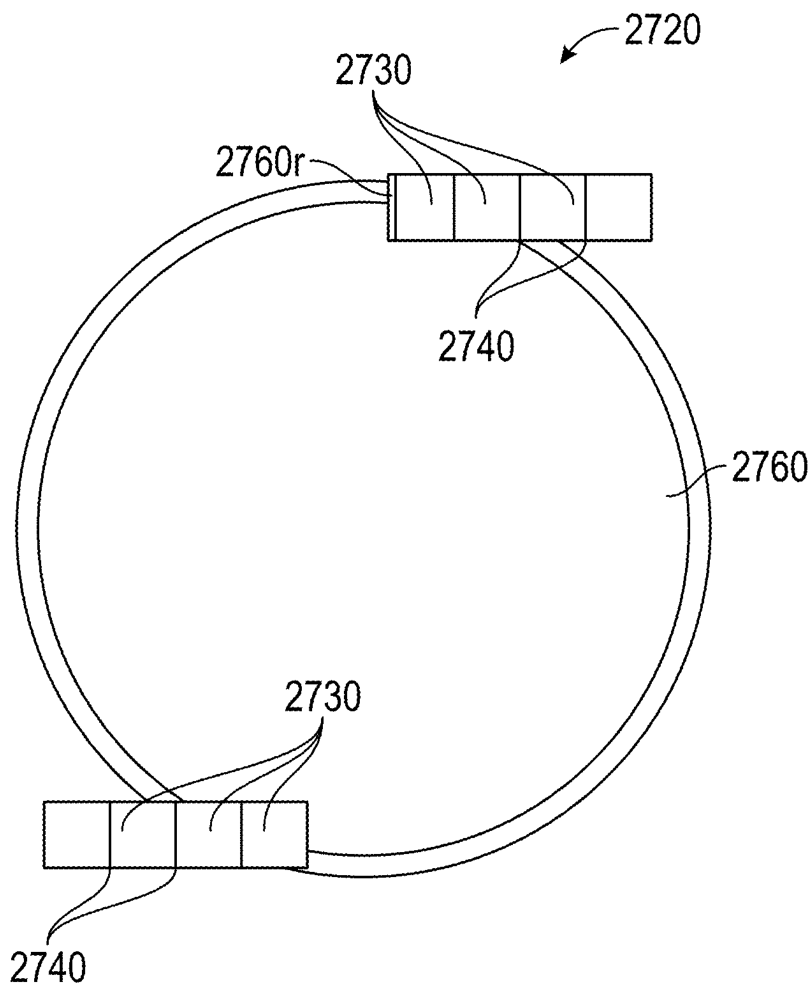


FIG. 8

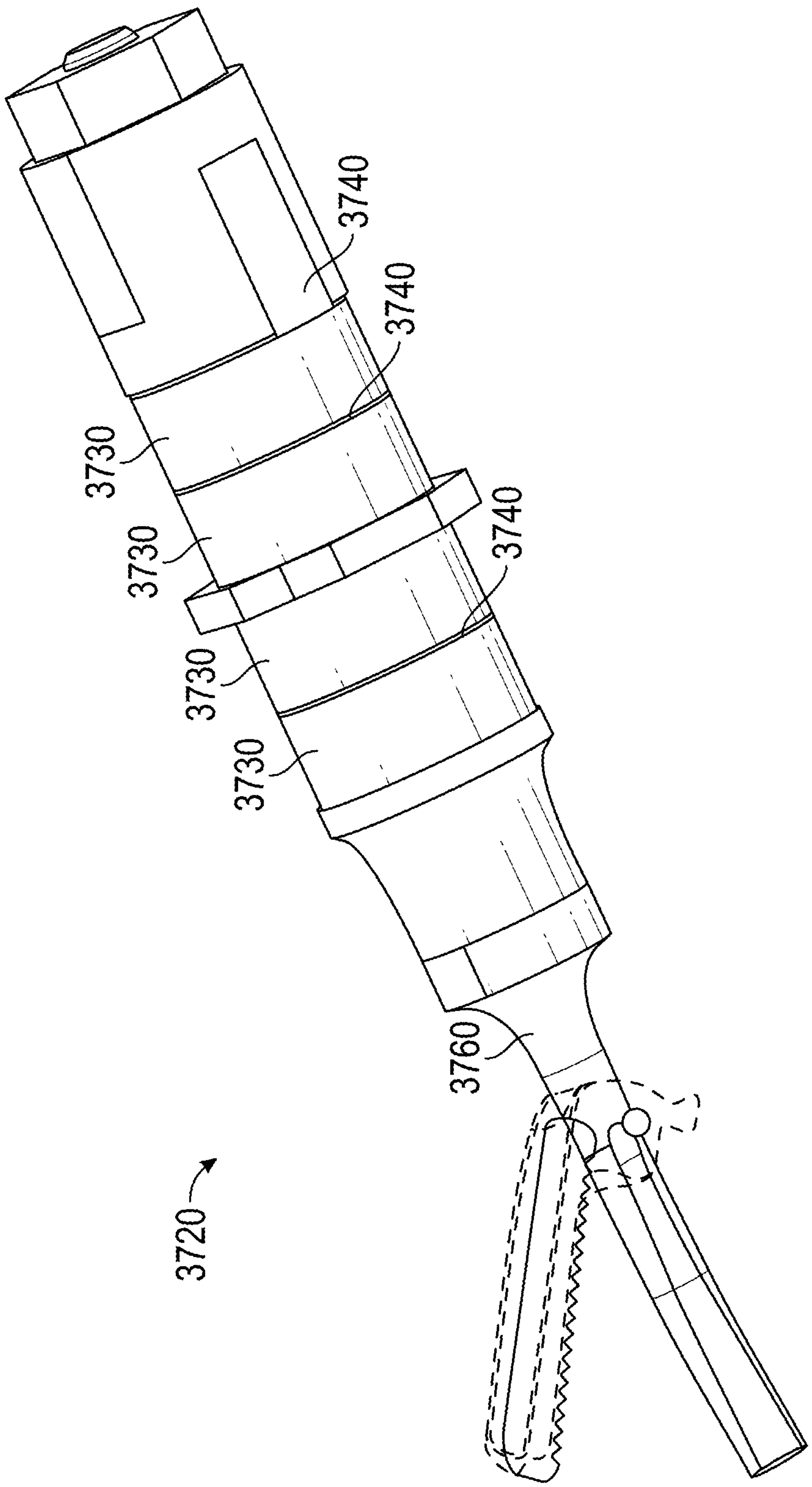


FIG. 9

ARTICULATING ULTRASONIC SURGICAL INSTRUMENTS AND SYSTEMS

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of U.S. patent application Ser. No. 17/797,175, filed Aug. 3, 2022, which is a U.S. National Stage Application filed under 35 U.S.C. § 371(a) of International Patent Application No. PCT/US2021/020473, filed Mar. 2, 2021, which claims the benefit of the filing date of provisional U.S. Patent Application No. 63/003,985, filed Apr. 2, 2020.

FIELD

[0002] The present disclosure relates to surgical instruments and systems and, more particularly, to articulating ultrasonic surgical instruments and systems including distally-located ultrasonic transducers.

BACKGROUND

[0003] Ultrasonic surgical instruments and systems utilize ultrasonic energy, i.e., ultrasonic vibrations, to treat tissue. More specifically, a typical ultrasonic surgical instrument or system includes a transducer configured to produce and transmit mechanical vibration energy at ultrasonic frequencies along a waveguide to an ultrasonic end effector configured to treat tissue, e.g., coagulate, cauterize, fuse, seal, cut, desiccate, fulgurate, or otherwise treat tissue. Traditionally, the transducer remains external of the surgical site, while the waveguide extends from the transducer into the surgical site to provide the ultrasonic energy to the ultrasonic end effector. The ultrasonic end effector is manipulated into position to treat a desired tissue or tissues.

[0004] Some ultrasonic surgical instruments and systems incorporate rotation features, thus enabling rotation of the ultrasonic end effector to a desired orientation within the surgical site. However, even in such instruments and systems, the ability to navigate within the surgical site via rotation and manipulation alone is limited.

SUMMARY

[0005] As used herein, the term "distal" refers to the portion that is being described which is further from a user, while the term "proximal" refers to the portion that is being described which is closer to a user. Further, to the extent consistent, any or all of the features of any of the devices detailed herein may be used in conjunction with any or all of the other devices detailed herein.

[0006] In accordance with aspects of the present disclosure, an articulating ultrasonic surgical end effector is provided. The end effector includes an articulation assembly, a clevis operably coupled to the articulation assembly, and a transducer assembly pivotably coupled to the clevis. The transducer assembly includes a transducer housing, and an ultrasonic transducer and a waveguide disposed within the transducer housing. The waveguide is operably coupled to the ultrasonic transducer and extends distally from the ultrasonic transducer. The transducer assembly also includes an ultrasonic blade and a clamp arm. The ultrasonic blade is disposed at the distal end of the waveguide, and extending from the transducer housing. The clamp arm is pivotably coupled to the transducer housing and movable relative to the ultrasonic blade between an open position and a clamping position. Ultrasonic energy produced by the ultrasonic transducer is transmitted along the waveguide to the ultrasonic blade for treating tissue.

[0007] In aspects of the present disclosure, a pulley and cable arrangement operably couples the transducer assembly to the clevis to permit pivoting of the transducer assembly relative to the clevis.

[0008] In aspects of the present disclosure, a pulley and cable arrangement extends between the clamp arm and the clevis to permit pivoting of the clamp arm relative to the ultrasonic blade regardless of an articulated position of the transducer assembly relative to the clevis.

[0009] In aspects of the present disclosure, the end effector further includes a pulley gear rotatably coupled to a proximal portion of the clevis, a distal gear rotatably coupled to a distal portion of the clevis and operably coupled to the transducer assembly such that rotation of the distal gear pivots the transducer assembly relative to the clevis, and a driver, belt, or chain coupling the pulley gear to the distal gear. The end effector may further include a first cable and a second cable operably coupled to the pulley gear. The first cable and the second cable may be configured to rotate the pulley gear to cause the transducer assembly to pivot relative to the clevis when one of the first cable or the second cable is proximally actuated, and articulate a distal articulation link of the articulation assembly relative to a proximal articulation link of the articulation assembly when both of the first cable and the second cable are proximally actuated.

[0010] In aspects of the present disclosure, the ultrasonic transducer includes at least one piezoelectric element and at least one electrode. The ultrasonic transducer, in aspects, defines a circular cross-sectional configuration.

[0011] In aspects of the present disclosure, a first notch and a second notch are defined along an outer surface of the ultrasonic transducer, on opposing sides thereof. The ultrasonic transducer may include at least one piezoelectric element operably coupled to the first notch and at least one piezoelectric element operably coupled to the second notch. The piezoelectric elements may be configured to produce ultrasonic vibrations in a torsional or longitudinal direction.

[0012] A surgical instrument provided in accordance with aspects of the present disclosure includes a handle assembly having an elongated body portion extending distally therefrom, and an articulating ultrasonic surgical end effector according to any of the above aspects, wherein the articulation assembly thereof extends distally from the elongated body portion of the handle assembly.

[0013] A surgical system provided in accordance with aspects of the present disclosure includes a robotic surgical system having a control device, a robotic arm, and an articulating ultrasonic surgical end effector according to any of the above aspects, wherein the articulation assembly extends distally from the robotic arm of the robotic surgical system.

[0014] In yet another aspect of the present disclosure, an articulating ultrasonic surgical end effector is provided and includes an articulating section having a plurality of articulating links configured to enable articulation in at least two different directions and a transducer assembly extending distally from the articulating section. The transducer assembly includes a transducer housing, and an ultrasonic transducer and a waveguide disposed within the transducer housing. The waveguide is operably coupled to, and extends distally from, the ultrasonic transducer. The transducer assembly further includes an ultrasonic blade disposed at the distal end of the waveguide and extending from the transducer housing. A clamp arm is pivotably coupled to the transducer housing and movable relative to the ultrasonic blade between an open position and a clamping position. Ultrasonic energy produced by the ultrasonic transducer is transmitted along the waveguide to the ultrasonic blade for treating tissue.

[0015] In aspects of the present disclosure, a plurality of articulation cables is operably coupled to at least one of the plurality of articulating links and configured to articulate the articulating section.

[0016] In aspects of the present disclosure, the ultrasonic transducer includes at least one piezoelectric element and at least one electrode.

[0017] In aspects of the present disclosure, the ultrasonic transducer defines a circular cross-sectional configuration.

[0018] In aspects of the present disclosure, a first notch and a second notch are defined along an outer surface of the ultrasonic transducer, on opposing sides thereof. The ultrasonic transducer may include at least one piezoelectric element operably coupled to the first notch and at least one piezoelectric element operably coupled to the second notch. The piezoelectric elements may be configured to produce ultrasonic vibrations in a torsional or longitudinal direction.

[0019] A surgical instrument provided in accordance with aspects of the present disclosure includes a handle assembly having an elongated body portion extending distally therefrom, and an articulating ultrasonic surgical end effector according to any of the above aspects, wherein the articulating section thereof extends distally from the elongated body portion of the handle assembly.

[0020] A surgical system provided in accordance with aspects of the present disclosure includes a robotic surgical system having a control device, a robotic arm, and an articulating ultrasonic surgical end effector according to any of the above aspects, wherein the articulation section thereof extends distally from the robotic arm of the robotic surgical system.

BRIEF DESCRIPTION OF THE DRAWINGS

[0021] The above and other aspects and features of the present disclosure will become more apparent in view of the following detailed description when taken in conjunction with the accompanying drawings wherein like reference numerals identify similar or identical elements and:

[0022] FIG. **1** is a side, perspective view of an endoscopic surgical instrument configured for use in accordance with the aspects and features of present disclosure and including an ultrasonic articulating end effector;

[0023] FIG. **2** is a schematic illustration of a robotic surgical system configured for use in accordance with the aspects and features of present disclosure and including an ultrasonic articulating end effector;

[0024] FIG. **3** is an enlarged, side, perspective view of a first side of an ultrasonic articulating end effector pivoted in a first direction in accordance with the present disclosure and configured for use with the endoscopic surgical instrument of FIG. **1**, the robotic surgical system of FIG. **2**, or any other suitable surgical instrument or system;

[0025] FIG. **4** is an enlarged, side, perspective view of a second side of the ultrasonic articulating end effector of FIG. **3** pivoted in the first direction;

[0026] FIG. **5** is an enlarged, side, perspective view of another ultrasonic articulating end effector pivoted in a first direction in accordance with the present disclosure and configured for use with the endoscopic surgical instrument of FIG. **1**, the robotic surgical system of FIG. **2**, or any other suitable surgical instrument or system;

[0027] FIG. **6** is an enlarged, side, perspective view of the ultrasonic articulating end effector of FIG. **5** pivoted in a second direction;

[0028] FIG. **7** is a side view of an exemplary drive mechanism usable with the end effector of FIG. **5**;

[0029] FIG. **8** is a cross-sectional view of a torsional ultrasonic transducer configured for use with the ultrasonic articulating end effector of FIGS. **3** or **5**, or any other suitable surgical instrument or system; and

[0030] FIG. **9** is a side, perspective, view of another ultrasonic transducer configured for use with the ultrasonic articulating end effector of FIGS. **3** or **5**, or any other suitable surgical instrument or system.

DETAILED DESCRIPTION

[0031] The present disclosure relates to articulating ultrasonic surgical instruments and systems that include distally-located ultrasonic transducers sized to enable use in minimally-invasive surgical procedures and/or other surgical procedures.

[0032] In some aspects, the ultrasonic transducer, waveguide, blade, and jaw are pivotable relative to a clevis to enable pitch articulation of the instrument, while a pivoting joint coupling the clevis to the shaft enables yaw articulation. By enabling pivoting of the ultrasonic transducer, waveguide, blade, and jaw relative to the clevis, the jaw and blade are positioned closer to the pivot point which creates less dead space, with the larger components, e.g., the ultrasonic transducer and portions of the waveguide, positioned in a "tail" region on the opposite side of the pivot. A housing enclosing the ultrasonic transducer and portions of the waveguide may include supports to retain the transducer/waveguide assembly at one or more node locations or other suitable location(s). The articulation in these aspects may be accomplished via a belt driven or chain driven configuration, cables, or in any other suitable manner and may be actuated by the rotation of a robotic motor, which may then translate through a series of gears and/or pulleys to cause the desired articulation.

[0033] In other aspects, a robotic "knuckle" or "wrist" is utilized. Unlike the aspects described above, the ultrasonic transducer is positioned distally of all of the articulation components. In such aspects, the transducer/waveguide assembly may likewise be mounted at a node, e.g., wherein a housing sandwiches a flange of the waveguide at a node, or other suitable location. In such aspects, the articulation may be accomplished via a cable driven system or other suitable mechanism. Combinations of the above aspects are also contemplated.

[0034] Referring generally to FIG. **1**, an endoscopic surgical instrument exemplifying the aspects and features of the present disclosure is shown generally identified by reference numeral **10**. For the purposes herein, endoscopic surgical instrument **10** is generally described. Aspects and features of endoscopic surgical instrument **10** not germane to the understanding of the present disclosure are omitted to avoid obscuring the aspects and features of the present disclosure in unnecessary detail.

[0035] Endoscopic surgical instrument **10** generally includes a handle assembly **12**, an elongated body portion **14**, and an articulating ultrasonic surgical end effector **200**. Aspects of end effector **200** and its components are described in greater detail below. Handle assembly **12** supports a battery assembly **18** and a generator assembly **20**, and includes a first rotation knob **22**, a second rotation knob **23**, an activation button **24**, and a clamp trigger **26**.

[0036] Clamp trigger **26** of endoscopic surgical instrument **10** is selectively manipulatable to actuate a motor, other powered drive mechanism, or a manual drive mechanism, e.g., gears, pulleys, tension cables, etc., to transition end effector **200** between an open condition and a clamping condition, as detailed below.

[0037] First rotation knob **22** is selectively manipulatable, in a first manner, e.g., a rotational manner, to rotate (e.g., roll) elongated body portion **14** and, thus, end effector **200** relative to handle assembly **12** (e.g., around a longitudinal axis "L" defined by elongated body portion **14** in either direction represented by arrow "R"). First rotation knob **22** is selectively manipulatable in a second manner, e.g., a translational manner, to articulate end effector **200** (e.g., yaw articulation) relative to elongated body portion **14** about a pivot axis "P" (FIGS. **3** and **4**) in one direction. First rotation knob **22** may be controlled by a manual drive mechanism or by a powered drive mechanism. Second rotation knob **23** is selectively manipulatable, in a first manner, e.g., a rotational manner, to actuate a motor, other powered drive mechanism, or a manual drive mechanism, e.g., gears, pulleys, tension cables, etc., to articulate (e.g., yaw articulation) end effector **200** relative to elongated body portion **14** about pivot axis "P" (FIGS. **3** and **4**), and in a second manner, e.g., a translational manner, to pivot (e.g., pitch articulation) components of end effector **200**, as detailed below. As an alternative to first and second rotation knobs **22**, **23**, additional or alternative suitable actuation mechanism(s), e.g., toggle switches, joysticks, buttons, etc., may be provided.

[0038] Battery assembly **18** and generator assembly **20** cooperate, upon activation of activation button **24**, to supply power to end effector **200** to enable the generation of ultrasonic energy for treating tissue therewith, e.g., to coagulate, cauterize, fuse, seal, cut, desiccate, fulgurate, or otherwise treat tissue, as detailed below. Battery assembly **18** and generator assembly **20** are each releasably secured to handle assembly **12**, and are removable therefrom to facilitate disposal of handle assembly **12**, with the exception of battery assembly **18** and generator **20**. However, it is contemplated that any or all of the components of endoscopic surgical instrument **10** may be configured as disposable single-use components or sterilizable multi-use components, and/or that endoscopic surgical instrument **10** may be connectable to a remote power source or generator rather than having such components on-board.

[0039] Referring generally to FIG. **2**, an aspect of a robotic surgical system exemplifying the aspects and features of the present disclosure is shown generally identified by reference numeral **1000**. For the purposes herein, robotic surgical system **1000** is generally described. Aspects and features of robotic surgical system **1000** not germane to the understanding of the present disclosure are omitted to avoid obscuring the aspects and features of the present disclosure in unnecessary detail.

[0040] Robotic surgical system **1000** generally includes a plurality of robot arms **1002**, **1003**; a control device **1004**; and an operating console **1005** coupled with control device **1004**. Operating console **1005** may include a display device **1006**, which may be set up in particular to display three-dimensional images; and manual input devices **1007**, **1008**, by means of which a person (not shown), for example a surgeon, may be able to telemanipulate robot arms **1002**, **1003** in a first operating mode. Robotic surgical system **1000** may be configured for use on a patient **1013** lying on a patient table **1012** to be treated in a minimally invasive manner. Robotic surgical system **1000** may further include a database **1014**, in particular coupled to control device **1004**, in which are stored, for example, pre-operative data from patient **1013** and/or anatomical atlases.

[0041] Each of the robot arms **1002**, **1003** may include a plurality of members, which are connected through joints, and an attaching device **1009**, **1011**, to which may be attached, for example, a surgical tool "ST" supporting an end effector **200**, **1100**. End effector **200**, as noted above with respect to endoscopic surgical instrument **10** (FIG. **1**), and as described in greater detail below, is an articulating ultrasonic surgical end effector. End effector **1100** may be any other suitable surgical end effector, e.g., an endoscopic camera, other surgical tool, etc. Robot arms **1002**, **1003** may be driven by electric drives, e.g., motors, that are connected to control device **1004**. Control device **1004** (e.g., a computer) may be configured to activate the motors, in particular by means of a computer program, in such a way that robot arms **1002**, **1003**, their attaching devices **1009**, **1011**, and, thus, the surgical tools "ST" (including end effectors **200**, **1100**) execute a desired movement and/or function according to a corresponding input from manual input devices **1007**, **1008**, respectively. Control device **1004** may also be configured in such a way that it regulates the movement of robot arms **1002**, **1003** and/or of the motors.

[0042] Turning to FIGS. **3** and **4**, articulating ultrasonic surgical end effector **200** includes an articulation assembly **201**, a clevis **212** extending distally from the articulation assembly **201**, and a transducer assembly **280** pivotable relative to the clevis **212**. Transducer assembly **280** includes a transducer housing **282** including a shaft **284** extending distally from a body of transducer housing **282**, an ultrasonic blade **286** extending distally from shaft **284**, and a clamp arm **288** pivotable relative to shaft **284** and ultrasonic blade **286** via a clamp pulley **283** or other suitable mechanism.

[0043] Articulation assembly **201** operably couples end effector **200** to a surgical instrument or system. Articulation assembly **201** includes a proximal articulation link **201*a*** and a distal articulation link **201*b*** pivotable relative to the proximal articulation link **201*a*** about a pivot axis "P" for achieving yaw articulation of the end effector **200**. Proximal articulation link **201*a*** is coupled to an elongated body portion **14** (FIG. **1**) or an attaching device **1009**, **1011** (FIG. **2**) of a handheld or robotic system and distal articulation link **201*b*** is operably coupled to clevis **212** of end effector **200**. Articulation of distal articulation link **201*b*** about pivot axis "P" effects corresponding articulation of clevis **212** extending therefrom in the same direction about pivot axis "P". As described below, a plurality of cables, for example, first cable **252**, second cable **254**, third cable **256**, and fourth cable **258**, extend through end effector **200** for manipulating end effector **200** and its components. Although articulation assembly **201** is described and illustrated as including a proximal articulation link **201*a*** and a distal articulation link **201*b***, articulation assembly **201** may include only a single link or more than two links. Additionally, although end effector **200** is illustrated and described as including an articulation assembly **201**, a proximal portion of end effector **200** (e.g., a proximal portion of clevis **212**) may be coupled directly to a shaft of a surgical instrument or robotic arm without the inclusion of any articulation assembly such as articulation assembly **201**.

[0044] Clevis **212** includes a pair of spaced-apart arms, in particular, right arm **212**R (FIG. **3**) and left arm **212**L (FIG. **4**) extending distally therefrom. Right arm **212**R and left arm **212**L are operably coupled to a portion of transducer assembly **280** (e.g., to shaft **284** via transducer housing **282**) and pivotably secure transducer assembly **280** between right arm **212**R and left arm **212**L.

[0045] With reference to FIG. **3**, a proximal portion of right arm **212**R houses a pulley gear **220** which is rotatably coupled to right arm **212**R via a pivot pin **220***p* and rotatable relative to right arm **212**R. First cable **252** and second cable **254** extend distally through articulation assembly **201** (e.g., through channels defined through articulation assembly **201**) and are routed around a pulley portion (not shown) of pulley gear **220** or operably coupled at their respective distal ends to the pulley portion of pulley gear **220**. A distal portion of right arm **212**R houses a distal gear **222** which is rotatably coupled to the distal portion of right arm **212**R via a pivot pin **222***p*. Distal gear **222** is operably coupled to a gear portion (not shown) of pulley gear **220** via a belt **221** such that rotation of pulley gear **220** effects corresponding rotation of distal gear **222**, though other coupling components may be used (e.g., geared, chain, etc.) for coupling pulley gear **220** to distal gear **222**. Distal gear **222** is secured to shaft **284** (or another portion of transducer assembly **280**) such that rotation of distal gear **222** effects rotation of shaft **284** relative to a longitudinal axis defined by clevis **212** to achieve pitch articulation.

[0046] According to the configuration described above, simultaneous proximal translation of first cable **252** and second cable **254** in the direction of arrow "A" pulls pulley gear **220** thereby urging distal articulation link **201***b* to rotate about pivot axis "P" in the direction of arrow "a" to achieve yaw articulation in one direction. Additionally, simultaneous proximal translation of first cable **252** and second cable **254** in the direction of arrow "A" may be met with distal translation or slack provided to third cable **256** and fourth cable **258** so as to not impart (or reduce) tension upon third cable **256** and fourth cable **258** when articulating in the direction of arrow "a". Proximal translation of only first cable **252** in the direction of arrow "A" (met with distal translation or slack provided to second cable **254**) causes shaft **284** to pivot relative to clevis **212** in the direction of arrow "b" to achieve pitch articulation in one direction. Conversely, proximal translation of only second cable **254** in the direction of arrow "A" (met with distal translation or slack provided to first cable **252**) causes shaft **284** to pivot relative to clevis **212** in the direction of arrow "c" to achieve pitch articulation in an opposite direction. Although described as two cables, first cable **252** and second cable **254** may be respective ends of a single cable.

[0047] Turning to FIG. **4**, left arm **212**L of clevis **212** houses a proximal pulley **224**, a middle pulley **226**, and a distal pulley **228**. In particular, proximal pulley **224** is rotatably coupled to left arm **212**L via a pivot pin **224***p*, middle pulley **226** is rotatably coupled to left arm **212**L via a pivot pin **226***p*, and distal pulley **228** is rotatably coupled to left arm **212**L via a pivot pin **228***p*. Third cable **256** extends distally through articulation assembly **201** (e.g., through channels defined through articulation assembly **201**) and is routed about proximal pulley **224**, routed about middle pulley **226**, routed about distal pulley **228**, and secured to clamp pulley **283**. Similarly, fourth cable **258** extends distally through articulation assembly **201** (e.g., through channels defined through articulation assembly **201**) and is routed about proximal pulley **224**, routed about middle pulley **226**, routed about distal pulley **228**, and secured to clamp pulley **283**. Although described as two cables, third cable **256** and fourth cable **258** may be respective ends of one single cable.

[0048] Clamp pulley **283** is engaged with clamp arm **288** and is rotatably coupled to the distal end of shaft **284** via a pivot pin **283***p* such that rotation of clamp pulley **283** in a first direction relative to shaft **284** pivots clamp arm **288** towards a clamping position, wherein clamp arm **288** is positioned adjacent ultrasonic blade **286** for clamping tissue therebetween, and such that rotation of clamp pulley **283** in a second, opposite direction relative to shaft **284** pivots clamp arm **288** towards an open positon, wherein clamp arm **288** is further-spaced from ultrasonic blade **286**.

[0049] According to the configuration described above, simultaneous proximal translation of third cable **256** and fourth cable **258** in the direction of arrow "D" pulls middle pulley **226** and causes distal articulation link **201***b* to rotate about pivot axis "P" in the direction of arrow "e" to achieve yaw articulation in one direction (e.g., a direction opposite to that of arrow "a"). Additionally, simultaneous proximal translation of third cable **256** and fourth cable **258** in the direction of arrow "D" may be met with distal translation or slack provided to first cable **252** and second cable **254** (FIG. **3**) so as to not impart (or to reduce) tension on first cable **252** and second cable **254** when articulating in the direction of arrow "e". Proximal translation of only third cable **256** in the direction of arrow "D" (met with distal translation or slack provided to fourth cable **258**) causes clamp arm **288** to pivot relative to ultrasonic blade **286** in the direction of arrow "f" towards the clamping position. Conversely, proximal translation of only fourth cable **258** in the direction of arrow "D" (met with distal translation or slack provided to third cable **258**) causes clamp arm **288** to pivot relative to ultrasonic blade **286** in the direction of arrow "g" towards the open position. Alternatively, third cable **256** and fourth cable **258** may be configured as a single cable secured about clamp pulley **283** and having its two ends extending proximally from clamp pulley **283**, about distal pulley **228**, middle pulley **226**, and proximal pulley **224**, proximally from clevis **212** and articulation assembly **201**. In some aspects, a portion of fourth cable **258** and/or the pulley **283** may be disposed within transducer housing **282** to protect fourth cable **258** and pulley **283**. As an alternative to third cable **256** and fourth cable **258** being operatively coupled to a clamp pulley **283**, third cable **256** and fourth cable **258** may be coupled directly to clamp arm **288** on opposing sides of a pivot pin **283***p* to cause clamp arm **288** to pivot relative to ultrasonic blade **286**.

[0050] Referring additionally to FIG. **1**, with respect to use of end effector **200** with endoscopic surgical instrument **10**, the proximal ends of first, second, third, and fourth cables **252**, **254**, **256**, **258** of end effector **200** extend proximally through elongated body portion **12** to a suitable control mechanism configured to actuate cables or otherwise control articulation and clamping of the components of end effector **200**. The proximal ends of first and second cables **252**, **254** are operably coupled to second rotation knob **23** (or the drive component associated therewith) such that rotation of second rotation knob **23** in a first direction pivots transducer assembly **280** relative to clevis **212** in a first direction, and such that rotation of second rotation knob **23** in a second direction pivots transducer assembly **280** relative to clevis **212** in a second direction to achieve pitch articulation (e.g., in the direction of arrows "b" and "c" in FIG. **3**). Such a configuration enables articulation of clamp arm **288** and ultrasonic blade **286** to a desired orientation relative to tissue to be treated. Additionally, with the proximal ends of first and second cables **252**, **254** coupled to second rotation knob **23**, proximal translation of second rotation knob **23** (e.g., pulling second rotation knob **23** away from handle assembly **12**) urges distal articulation link **201***b* and clevis **212** to rotate about pivot axis "P" (e.g., in the direction of arrow "a") to achieve yaw articulation.

[0051] Proximal portions of third and fourth cables **256**, **258** are operably coupled to clamp trigger **26** such that actuation of clamp trigger **26** from an un-actuated position to an actuated position pivots clamp arm **288** relative to ultrasonic blade **286** from the open position to the clamping position (e.g., in the direction of arrow "f"). Conversely, proximal portions of third and fourth cables **256**, **258** are operably coupled to clamp trigger **26** such that return of clamp trigger **26** from the actuated position back to the un-actuated position pivots clamp arm **288** relative to ultrasonic blade **286** from the clamping position back to the open position (e.g., in the direction of arrow "g").

[0052] In addition to being coupled to clamp trigger **26**, proximal portions of third and fourth cables **256**, **258** may be operably coupled to first rotation knob **22** or other suitable component for controlling yaw articulation. For example, proximal portions of third and fourth cables **256**, **258** may be routed through a pulley system (not shown) or otherwise operably routed through clamp trigger **26**, with their respective proximal-most ends operably coupled to first rotation knob **22**. Alternatively, proximal portions of third and fourth cables **256**, **258** may be routed through a pulley system (not shown) or otherwise operably routed through first rotation knob **22**, with their respective proximal-most ends operably coupled to clamp trigger **26**. In either configuration, third and fourth cables **256**, **258** are operably coupled to first rotation knob **22** or other mechanism such that proximal translation of first rotation knob **22** (e.g., urging first rotation knob **22** or other mechanism proximally toward second rotation knob **23**) urges distal articulation link **201***b* and clevis **212** to rotate about pivot axis "P" (e.g., in the direction of arrow "e") to achieve yaw articulation in one direction.

[0053] Referring to FIGS. **2-4**, with respect to use of end effector **200** with robotic surgical system **1000**, the proximal ends of first, second, third, and fourth cables **252**, **254**, **256** **258** of end effector **200** extend proximally through robot arm **1002** and each operably couple to a corresponding motor of control device **1004**. Control device **1004** is operable, depending upon the input instructions received, to drive the appropriate motor thereof to pull the corresponding cable **252**, **254**, **256** **258** to pivot transducer assembly **280** relative to clevis **212** in a desired direction to position clamp arm **288** and ultrasonic blade **286** to a desired orientation relative to tissue to be treated for achieving pitch articulation, to articulate end effector **200** about pivot axis "P" for achieving yaw articulation, to rotate, e.g., roll, end effector **200** about longitudinal axis "L," and to pivot clamp arm **288** between the open and clamping positions to clamp tissue between clamp arm **288** and ultrasonic blade **286**.

[0054] Turning back to FIGS. **3** and **4**, an inner assembly **270** of transducer assembly **280** is disposed partially within transducer housing **282**, extends through shaft **284**, and extends distally from shaft **284**. Inner assembly **270** includes an ultrasonic transducer **272**, which may be formed from a stack of piezoelectric transducer elements **273** or otherwise configured, an ultrasonic horn (not shown), and a waveguide **276**. Electrodes **274** are interdisposed between piezoelectric transducer elements **273** are electrically coupled to a source of energy, e.g., via lead wires **275** extending through the pivots that couple transducer assembly **280** with clevis **212** and proximally through the instrument or system to a source of energy. Upon energization of electrodes **274**, e.g., in response to activation of activation button **24** of surgical instrument (FIG. **1**) or in response to an appropriate instruction provided by control device **1004** of robotic surgical system **1000** (FIG. **2**), piezoelectric transducer elements **273** produce ultrasonic energy that is transmitted from ultrasonic transducer **272**, to the ultrasonic horn, and along waveguide **276**, which extends distally from transducer housing **282**, to ultrasonic blade **286**. Ultrasonic blade **286** extends distally from waveguide **276** and distally from shaft **284**. Ultrasonic blade **286** is positioned adjacent clamp arm **288** to enable clamping of tissue between ultrasonic blade **286** and clamp arm **288** in the clamping position of clamp arm **288**. Ultrasonic energy transmitted along waveguide **276** to ultrasonic blade **286** is communicated to tissue clamped between clamp arm **288** and ultrasonic blade **286** to treat tissue. Alternatively, clampless tissue treatment using just ultrasonic blade **286**, e.g., via plunging, scoring, etc., may also be performed.

[0055] Turning to FIGS. **5** and **6**, another exemplary articulating ultrasonic surgical end effector is described. Articulating ultrasonic surgical end effector **300** includes an articulating section **360** that operably couples end effector **300** to a surgical instrument or system and articulates a distal end of the end effector **300** relative to a longitudinal axis of the instrument. For example, articulating section **360** may be defined at the distal end of elongated body portion **14** of endoscopic surgical instrument **10** (FIG. **1**), at the distal end of attaching device **1009** of robot arm **1002** of robotic surgical system **1000** (FIG. **2**), or at any other suitable location for enabling use of end effector **300** with a corresponding surgical instrument or system.

[0056] Articulating section **360** includes a plurality of intermediate articulating components, e.g., links, joints, etc., including a proximal articulating joint **373***a*, having a proximal link **372***a* and a distal link **374***a*, and a distal articulating joint **373***b*, having a proximal link **372***b* and a distal link **374***b*. The proximal link **372***a* of the proximal articulating joint **373***a* is rotatable (e.g., pivotable) relative to the distal link **374***a* of the proximal articulating joint **373***a*. Similarly, the proximal link **372***b* of the distal articulating joint **373***b* is rotatable (e.g., pivotable) relative to the distal link **374***b* of the distal articulating joint **373***b*. The distal link **374***a* of the proximal articulating joint **373***a* is fixedly coupled to the proximal link **372***b* of the distal articulating joint **373***b*. A plurality of articulation cables **380**, e.g., four (4) articulation cables, or other suitable actuators, extend through articulating section **360** at different radial orientations.

[0057] In one aspect, articulation cables **380** are operably coupled to distal link **374***b* of distal articulation joint **373***b* at the distal ends thereof and extend proximally from the distal link **374***b*, through articulating section **360**, and to an articulation sub-assembly (not shown), e.g., a gearbox assembly, actuator assembly, etc., to enable selective articulation of the distal link **374***b* of the distal articulating joint **373***b* (and, thus the distal end of end effector **300**) relative to the proximal link **372***a* of proximal articulating joint **373***a*, e.g., about at least two axes of articulation (yaw and pitch articulation, for example).

[0058] Alternatively, one pair of articulation cables **380** may be operably coupled to a distal link **374***b* of distal articulating joint **373***b* while a second pair of articulation cables **380** is operably coupled to the distal link **374***a* of proximal articulating joint **373***a*. In this configuration, one pair of articulation cables **380** is operably coupled to the distal link **374***b* of distal articulating joint **373***b* at the distal ends thereof and extend proximally from the distal link **374***b*, through articulating section **360**, and to an articulation sub-assembly (not shown), e.g., a gearbox assembly, actuator assembly, etc., to enable selective articulation of the distal link **374***b* (and, thus the distal end of end effector **300**) relative to the proximal link **372***b* of the distal articulating joint **373***b*. The second pair of articulation cables **380** is operably coupled to a distal link **374***a* of proximal articulating joint **373***a* at the distal ends thereof and extend proximally from the distal link **374***a*, through articulating section **360**, and to an articulation sub-assembly (not shown), e.g., a gearbox assembly, actuator assembly, etc., to enable selective articulation of the distal link **374***a* relative to the proximal link **372***a* of the proximal articulating joint **373***a* (and, thus the distal end of end effector **300**). At least two of the links are oriented differently and/or differently configured to define different pivot axes, thereby enabling the articulation in at least two different directions. Articulation cables **380** are arranged to define a generally square configuration, although other suitable configurations are also contemplated.

[0059] With respect to articulation of the end effector **300** relative to the proximal link **372**, actuation of articulation cables **380** is effected in pairs. More specifically, in order to pitch the end effector **300**, the upper pair of articulation cables **380** is actuated in a similar manner while the lower pair of articulation cables **380** is actuated in a similar manner relative to one another but an opposite manner relative to the upper pair of articulation cables **380**. With respect to yaw articulation, the right pair of articulation cables **380** is actuated in a similar manner while the left pair of articulation cables **380** is actuated in a similar manner relative to one another but an opposite manner relative to the right pair of articulation cables **380**.

[0060] The distal link **374***b* of the distal articulating joint **373***b* is operably coupled to a proximal portion of the transducer housing **430** such that movement and articulation of the proximal articulating joint **373***a* and/or distal articulating joint **373***b* effects corresponding movement and articulation of the transducer housing **430**.

[0061] End effector **300** further includes a shaft **440**, forming a distal extension portion of transducer housing **430**, a clamp arm **450** pivotable relative to shaft **440**, a clamp pulley **460** operably coupled to clamp arm **450**, and an inner assembly **470** disposed partially within transducer housing **430**, extending through shaft **440**, and extending distally from shaft **440**. A proximal end of transducer housing **430** is fixedly mounted to a distal portion of articulating section **360** (e.g., at distal link **374**). Shaft **440** includes two proximally extending arms **412** which couple the shaft **440** to the body of transducer housing **430**.

[0062] Clamp pulley **460** is engaged with clamp arm **450** and rotatably coupled to the distal end of shaft **440** such that rotation of clamp pulley **460** in a first direction relative to shaft **440** pivots clamp arm **450** towards a clamping position, wherein clamp arm **450** is positioned adjacent ultrasonic blade **478** for clamping tissue therebetween, and such that rotation of clamp pulley **460** in a second, opposite direction relative to shaft **440** pivots clamp arm **450** towards an open positon, wherein clamp arm **450** is further-spaced from ultrasonic blade **478**.

[0063] In an aspect, clamp pulley **460** is operably coupled to one or more of articulating cables **380** such that actuation of one or more of articulating cables **380** effects rotation of clamp pulley **460** and in turn clamp arm **450**. Alternatively, referring now to FIG. **7** a clamp drive mechanism **3900** including independent clamp drive cables **390** or a single drive rod (not shown) may extend through articulating section **360** to couple to, and control rotation of, a clamp lever **392** to pivot clamp arm **450** relative to ultrasonic blade **478**. Such a configuration may use a camming mechanism to pivot clamp arm **450** relative to ultrasonic blade **478** between the open and clamping positions. In particular, distal ends of two drive cables **390** are coupled to the clamp lever **392** on opposite sides of a pivot point **392***p* to pivot the clamp lever **392** about the pivot point **392***p* via control of the drive cables **390**. A pair of distal drive cables **394** extend distally from the clamp lever **392** to the clamp arm **450**. In particular, proximal ends of the distal drive cables **394** are coupled to the clamp lever **392** and distal ends of the distal drive cables **394** are coupled to the clamp arm **450** on opposite sides of a pivot point **450***p* to control the pivoting of clamp arm **450** relative to the ultrasonic blade **478**.

[0064] Referring to FIGS. **2**, **5**, and **6**, with respect to use of end effector **300** with robotic surgical system **1000**, the proximal ends of articulating cables **380** extend proximally through robot arm **1002** and each operably couple to a corresponding motor of control device **1004**. Control device **1004** is operable, depending upon the input instructions received, to drive the appropriate motor thereof to pull the corresponding articulating cable **380** to articulate transducer housing **430** in a desired direction to position clamp arm **450** and ultrasonic blade **478** to a desired orientation relative to tissue to be treated, to pivot clamp arm **450** between the open and clamping positions to clamp tissue between clamp arm **450** and ultrasonic blade **478**, and/or to rotate end effector **300** about its longitudinal axis.

[0065] Inner assembly **470** is disposed partially within transducer housing **430**, extends through shaft **440**, and extends distally from shaft **440**. Inner assembly **470** includes an ultrasonic transducer **472**, which may be formed from a stack of piezoelectric elements **473** or otherwise configured, an ultrasonic horn (not shown), and a waveguide **476**. Electrodes **474** are interdisposed between piezoelectric transducer elements **473** and are electrically coupled to a source of energy, e.g., via lead wires **475** extending through the articulating section **360** and proximally through the instrument or system to the source of energy. Upon energization of electrodes **474**, e.g., in response to activation of activation button **26** of surgical instrument (FIG. **1**) or in response to an appropriate instruction provided by control device **1004** of robotic surgical system **1000** (FIG. **2**), piezoelectric elements **473** produce ultrasonic energy that is transmitted from ultrasonic transducer **472**, to the ultrasonic horn (not shown), and along waveguide **476**, which extends distally from transducer housing **430**, to ultrasonic blade **478**. Ultrasonic blade **478** extends distally from waveguide **476** and distally from shaft **440**. Ultrasonic blade **478** is positioned adjacent clamp arm **450** to enable clamping of tissue between ultrasonic blade **478** and clamp arm **450** in the clamping position of clamp arm **450**. Ultrasonic energy transmitted along waveguide **476** to ultrasonic blade **478** is communicated to tissue clamped between clamp arm **450** and ultrasonic blade **478** to treat tissue. Alternatively, clampless tissue treatment using just ultrasonic blade **478**, e.g., via plunging, scoring, etc., may also be performed.

[0066] Turning now to FIG. **8**, a cross-section view of another ultrasonic transducer usable with end effector **200** or end effector **300** is shown as a torsional ultrasonic transducer **2720**. Torsional ultrasonic transducer **2720** may be disposed in transducer housing **282** of end effector **200** (FIGS. **3** and **4**) or transducer housing **430** of end effector **300** (FIGS. **5** and **6**), for example. Torsional ultrasonic transducer **2720** generates torsional ultrasonic vibrations in a circumferential direction of the piezoelectric transducer elements **2730** which are transmitted distally along the waveguide **2760** to the ultrasonic blade **286** (FIG. **3**) or ultrasonic blade **478** (FIG. **5**).

[0067] Each piezoelectric transducer element **2730** is operably coupled to an opposite side of waveguide **2760** to generate the torsional vibration in the circumferential direction of the piezoelectric transducer elements **2730**. Any number of piezoelectric transducer elements **2730** may be utilized including a single or multiple stacked elements. In an aspect, each piezoelectric transducer element **2730** is disposed in a notch or recess **2760*r*** defined by the waveguide **2760**. Each lead wire **275** (FIG. **4**) or lead wire **475** (FIG. **6**) is coupled to a respective electrode **2740** which is coupled to respective piezoelectric transducer elements **2730** such that energizing of each electrode **2740** causes longitudinal vibration of the respective piezoelectric transducer elements **2730** and induces torsional vibration of the waveguide **2760** due to the positions being opposed to one another on opposite tangential sides of the waveguide **2760**. These torsional ultrasonic vibrations are imparted to waveguide **2760** which, in turn, transmits the torsional ultrasonic vibrations along waveguide **2760** to ultrasonic blade **286** (FIG. **3**) or ultrasonic blade **478** (FIG. **5**) for treating tissue therewith. It is contemplated that use of a torsional ultrasonic transducer, e.g., torsional ultrasonic transducer **2720**, may enable a reduction in the overall length of the transducer assembly, which may be advantageous in configurations where the transducer assembly is distally-positioned such as provided herein with respect to end effectors **200** (FIGS. **3** and **4**) and **300** (FIGS. **5** and **6**).

[0068] Turning now to FIG. **9**, a side, perspective view of another ultrasonic transducer usable with end effector **200** or end effector **300** is shown as a longitudinal ultrasonic transducer **3720**. Longitudinal ultrasonic transducer **3720** may be disposed in transducer housing **282** of end effector **200** (FIGS. **3** and **4**) or transducer housing **430** of end effector **300** (FIGS. **5** and **6**), for example. Longitudinal ultrasonic transducer **3720** generates longitudinal ultrasonic vibrations in a longitudinal direction of the piezoelectric transducer elements **3730** which are transmitted distally along the waveguide **3760** to the ultrasonic blade **286** (FIG. **3**) or ultrasonic blade **478** (FIG. **5**).

[0069] Any number of piezoelectric transducer elements **3730** may be utilized including a single or multiple stacked elements. In an aspect, at least one electrode **3740** is disposed between adjacent piezoelectric transducer elements **3730**. Each lead wire **275** (FIG. **4**) or lead wire **475** (FIG. **6**) is coupled to a respective electrode **3740** which is coupled to respective piezoelectric transducer elements **3730** such that energizing of each electrode **3740** causes longitudinal vibration of the respective piezoelectric transducer elements **3730** and induces longitudinal vibration of the waveguide **3760**. These longitudinal ultrasonic vibrations are imparted to waveguide **3760** which, in turn, transmits the ultrasonic vibrations along waveguide **3760** to ultrasonic blade **286** (FIG. **3**) or ultrasonic blade **478** (FIG. **5**) for treating tissue therewith.

[0070] While several specific versions of devices in accordance with the present disclosure are shown in the drawings, it is not intended that the disclosure be limited thereto, as it is intended that the disclosure be as broad in scope as the art will allow and that the specification be read likewise. Therefore, the above description should not be construed as limiting, but merely as exemplifications of particular aspects. Those skilled in the art will envision other modifications within the scope and spirit of the claims appended hereto.

[0071] It should be understood that various features of the devices disclosed herein may be combined in different combinations than the combinations specifically presented in the description and accompanying drawings. It should also be understood that, depending on the example, certain acts or events of any of the processes or methods described herein may be performed in a different sequence, may be added, merged, or left out altogether (e.g., all described acts or events may not be necessary to carry out the techniques). In addition, while certain features of devices in accordance with the present disclosure are described as being performed by a single module or unit for purposes of clarity, it should be understood that the techniques of this disclosure may be performed by a combination of units or modules associated with, for example, a medical device.

What is claimed is:

1. An articulating electrosurgical system, comprising:

an elongate body;

an articulation assembly coupled to a distal end of the elongate body and articulable in a first plane;

a support structure coupled to the articulation assembly; and

an ultrasonic assembly coupled to and articulable with respect to the support structure in a second plane, the second plane being different than the first plane, the ultrasonic assembly comprising:

an ultrasonic transducer; and

an ultrasonic blade operatively coupled to the ultrasonic transducer such that ultrasonic energy produced by the ultrasonic transducer is transmitted to the ultrasonic blade for treating tissue.

2. The articulating electrosurgical system of claim **1**, wherein articulation of the articulation assembly in the first plane provides yaw articulation for the ultrasonic assembly.

3. The articulating electrosurgical system of claim **1**, wherein articulation of the ultrasonic assembly in the second plane provides pitch articulation for the ultrasonic assembly.

4. The articulating electrosurgical system of claim **1**, further comprising a handle coupled to a proximal end of the elongate body, wherein the elongate body is rotatable with respect to the handle, and wherein rotation of the elongate body provides roll articulation for the ultrasonic assembly.

5. The articulating electrosurgical system of claim **1**, further comprising a robotic arm, wherein the elongate body is rotatable with respect to the robotic arm, and wherein rotation of the elongate body provides roll articulation for the ultrasonic assembly.

6. The articulating electrosurgical system of claim **1**, wherein the support structure comprises a clevis.

7. The articulating electrosurgical system of claim **1**, wherein the articulation assembly comprises a plurality of links.

8. The articulating electrosurgical system of claim **1**, wherein the ultrasonic assembly further comprises a clamp arm that is pivotable relative to the ultrasonic blade between an open position and a clamping position.

9. An articulating electrosurgical instrument, comprising:
- an instrument body;
- an articulating section coupled to the instrument body; and
- an ultrasonic assembly coupled to the articulating section, the ultrasonic assembly articulable with respect to the instrument body via the articulating section in at least a first plane and a second plane, the second plane being different from the first plane, the ultrasonic assembly including:
  - an ultrasonic transducer; and
  - an ultrasonic blade operatively coupled to the ultrasonic transducer such that ultrasonic energy produced by the ultrasonic transducer is transmitted to the ultrasonic blade for treating tissue.

10. The articulating electrosurgical instrument of claim **9**, wherein the instrument body comprises:
- a handle; and
- an elongate body comprising a proximal end coupled to the handle and a distal end coupled to the articulating section, wherein articulation in the first plane provides yaw articulation for the ultrasonic assembly with respect to the handle.

11. The articulating electrosurgical instrument of claim **10**, wherein articulation in the second plane provides pitch articulation for the ultrasonic assembly with respect to the handle.

12. The articulating electrosurgical instrument of claim **10**, wherein the elongate body is rotatable with respect to the handle, and wherein rotation of the elongate body provides roll articulation for the ultrasonic assembly with respect to the handle.

13. The articulating electrosurgical instrument of claim **9**, wherein the articulating section comprises at least one link.

14. The articulating electrosurgical instrument of claim **9**, wherein the articulating section comprises a support structure, and wherein the ultrasonic assembly is pivotably coupled to the support structure.

15. The articulating electrosurgical instrument of claim **14**, wherein the support structure comprises a clevis.

16. The articulating electrosurgical instrument of claim **9**, the ultrasonic assembly further comprising a clamp arm that is pivotable relative to the ultrasonic blade between an open position and a clamping position.

17. A robotic surgical system, comprising:
- a robotic arm; and
- the articulating electrosurgical instrument of claim **9** coupled to the robotic arm.

18. The robotic surgical system of claim **17**, wherein articulation in the first plane provides yaw articulation for the ultrasonic assembly with respect to the robotic arm.

19. The robotic surgical system of claim **17**, wherein articulation in the second plane provides pitch articulation for the ultrasonic assembly with respect to the robotic arm.

20. The robotic surgical system of claim **17**, wherein the instrument body is rotatable with respect to the robotic arm, and wherein rotation of the instrument body provides roll articulation for the ultrasonic assembly with respect to the robotic arm.

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